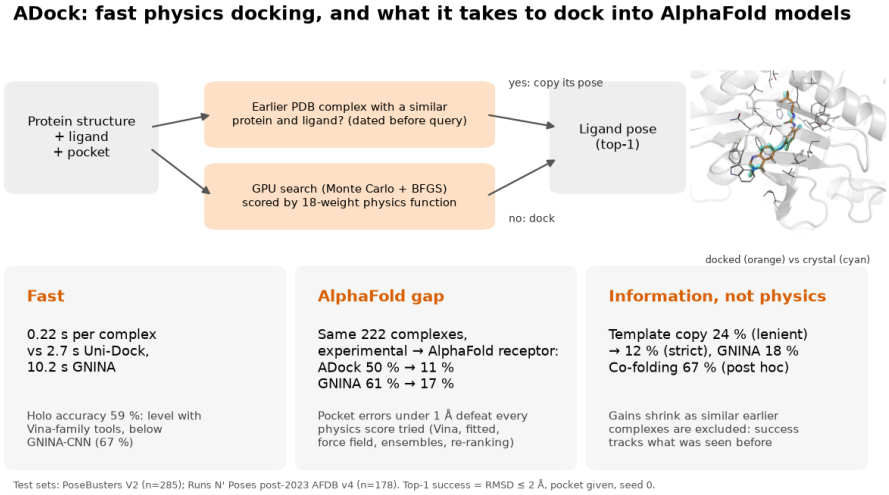
ADock — briefing

October 2026. AI-agent research project (24 h). Checked by independent agents, not yet by human experts. Details: `technical_report.pdf` *(4 pages) and* `docs/REPORT.md` *(full record).*

What was built

ADock is a fast docking program. Given a protein structure, a small molecule and the location of the binding pocket, it predicts the 3D pose in which the molecule binds.

- **On experimental structures** it runs a GPU version of the classical AutoDock Vina search, scored with a refitted, interpretable physics-style function.
- **On AlphaFold-predicted structures** it first checks whether an earlier PDB complex of a similar protein and ligand exists. If one does, ADock copies that ligand's pose; otherwise it docks.



How it was tested

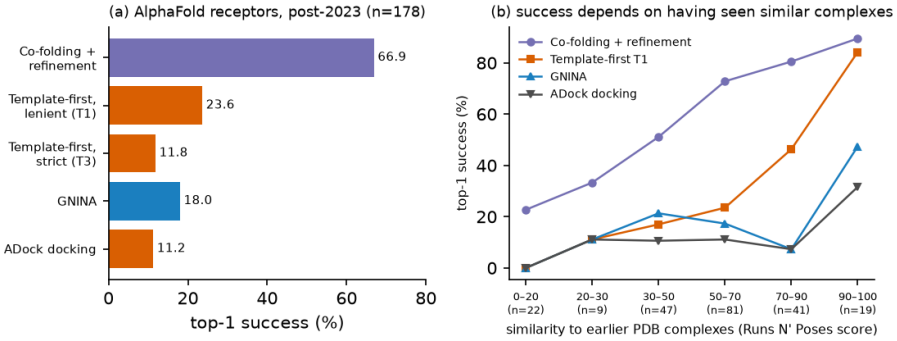
- **Success measure.** A prediction counts as correct if the top-ranked pose lies within 2 Å (RMSD) of the crystal pose. The pocket is given to every method in the same way.
- **Test sets.** These were held out until one final run:
  - **PoseBusters V2:** 285 complexes with experimental receptors.
  - **Runs N' Poses:** 222 complexes released after mid-2023, docked into both the experimental structure and the AlphaFold model (178 have the older AlphaFold database model used for the main AlphaFold claim).
- **Comparison tools:** AutoDock Vina, smina, Uni-Dock (GPU Vina) and GNINA, which adds neural-network rescoring and is the strongest open tool.
- **Pre-registration.** Three main claims and their statistical tests were written down before the test data were opened.

Main findings

1. **Fast, with standard accuracy.**
   - ADock takes 0.2 s per complex, against 2.7 s for Uni-Dock and 10 s for GNINA.
   - On experimental structures it is as accurate as the classical tools: 59 % vs Vina 55 % and GNINA without its network 59 %.
   - It stays below GNINA with neural-network rescoring, at 67 %.
2. **AlphaFold models break physics-based docking.**
   - On the same 222 complexes, ADock drops from 50 % on experimental to 11 % on AlphaFold receptors; GNINA drops from 61 % to 17 %.
   - The AlphaFold pockets are close to the real ones (within about half an ångström), but these small errors mislead every physics score we tried. Seven physics-side fixes gave no real gain (one apparent gain was a benchmark artefact). These were softer receptors, side-chain rebuilding, structure ensembles, a full force field, an AlphaFold-specific score, consensus scoring and re-ranking.
3. **Information from earlier complexes helps; physics does not.**

- Copying the pose from an earlier similar complex lifts AlphaFold success to 24 %, above GNINA's 18 %, but not significantly at the pre-specified level (p 0.08).
- When close relatives are excluded, the gain disappears (12 %).
- A co-folding model (Boltz-2), tried after the test, reached 67 %. Its success, however, falls to 23 % on the targets least similar to earlier PDB complexes.
- Success on AlphaFold receptors therefore mostly reflects what has been seen before.

4. **None of the three pre-registered main claims passed.**
   - ADock is not better than GNINA on AlphaFold receptors.
   - It is faster but less accurate than GNINA.
   - Its non-inferiority to the best physics tool was not significant after correction for testing three claims.



Success on AlphaFold receptors (post-2023 complexes). Left: all methods. Right: the same methods split by how similar each test complex is to complexes in the PDB before it. Co-folding was added after the test and is exploratory.

What this means

- **Experimental or high-confidence structures.** ADock is a practical choice for large-scale screening where speed matters: about 66 pockets per second per GPU node, or roughly 4 node-hours per million pockets (a projection). Where per-complex accuracy matters most, GNINA is better.
- **AlphaFold models of proteins with no similar solved complex.** No method tested here docks reliably (about 10–20 % success). Closing this gap needs pocket models accurate to well below 1 Å, or a score that tolerates such errors. Neither exists yet.
- **Research practice.** In nearly every experiment, gains seen during method selection shrank or vanished when checked on held-back data. Pre-registration and a held-out test changed what could be claimed.

Main limitations

- The binding pocket must be known in advance.
- Timings come from a node shared with other users, so treat them as upper bounds.
- Template thresholds, the improved "v3" settings and co-folding were chosen or added after seeing data, so those results are exploratory.
- The project was run by AI agents in 24 hours. Humans have not yet reviewed the code or results.